

DTSA 5728

Bayesian Statistical Modeling for Data Science Applications

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Elective	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective	Machine Learning Engineer (50.5 / 100)	38.9 / 100
SPECIALIZATION Bayesian Statistics for Data Science		

39

CURRENT AVG SCORE
across 6 job roles

67

POTENTIAL SCORE
with recommended additions

+37

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

This course introduces students to applied Bayesian statistical modeling. Topics include Bayesian linear regression, Bayesian generalized linear models, and an introduction to Bayesian hierarchal modeling. Special emphasis will be placed on the application of maximum a posterior estimation and Markov chain Monte Carlo algorithms on Bayesian statistical models. This course also demonstrate Bayesian statistical modeling in the Stan computing environment.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE		PROJECTED SCORE		POINT GAIN	PRIORITY
Machine Learning Engineer		50.5		77.8	+27.3	MEDIUM
ML Engineer		42.6		62.2	+19.6	MEDIUM
Data Scientist		40.0		77.0	+37.1	HIGH
AI Engineer		38.5		63.4	+24.9	MEDIUM
Applied Scientist		35.8		67.0	+31.2	HIGH
MLOps Engineer		26.3		51.8	+25.5	MEDIUM

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design

4 of 6 roles

systems

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1

HIGH

Add System Design coverage — required vocabulary in 4 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

GAP KEYWORDS BY ROLE

Machine Learning Engineer	50.5 → 77.8 (+27.3 pts)
data predictive sets learning tools exploratory design development	
ML Engineer	42.6 → 62.2 (+19.6 pts)
data learning machine systems model problems global tools	
Data Scientist	40.0 → 77.0 (+37.1 pts)
data sets behaviors social problems	
AI Engineer	38.5 → 63.4 (+24.9 pts)
following data learning model systems problems machine design	
Applied Scientist	35.8 → 67.0 (+31.2 pts)
computational research problems data learning systems industry design	
MLOps Engineer	26.3 → 51.8 (+25.5 pts)
data model systems learning machine tools development	

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum–Industry Alignment project team.